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Model Tuning

So till now, we've learned how to build machine learning models — but there's one big truth about models:
The first version of your model is never the best version.
That's where Model Tuning comes in.

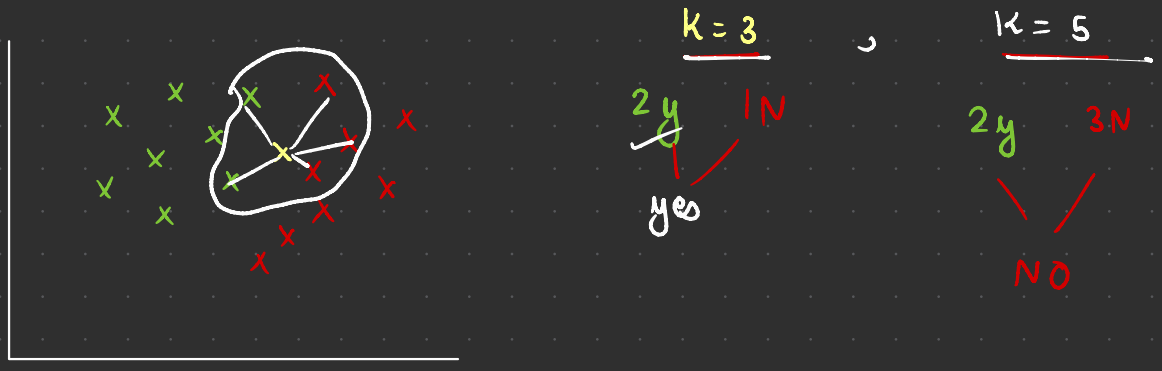
When we train a model, it has certain settings — these are called hyperparameters. These hyperparameters decide how your model behaves — like how deep should a tree be, how many neighbors in KNN, or what learning rate to use.

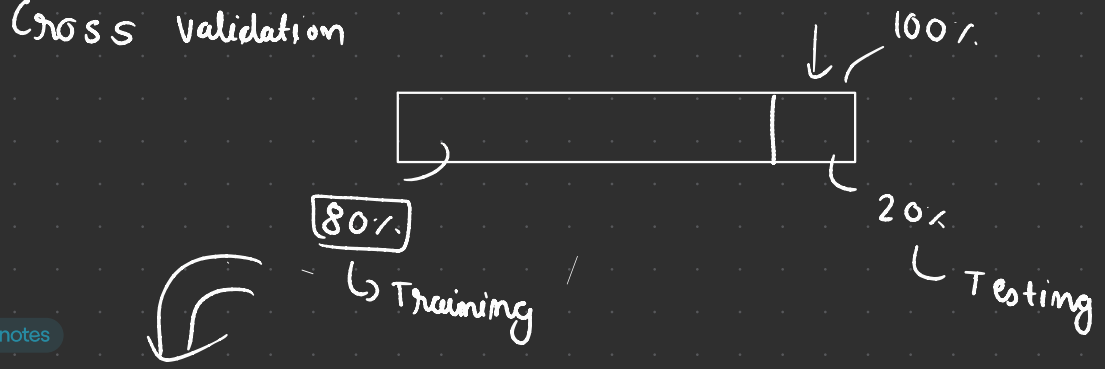
But if we just use random hyperparameters — our model might underfit, or overfit, or simply perform poorly.

So the goal of Model Tuning is:
1) To find the best combination of hyperparameters
2) So that our model performs better on new, unseen data — not just on the training data.

In short — tuning helps you squeeze out the best possible performance from your model — and that's why it's an essential step before finalizing any ML model. And this is also called hyper parameter tuning.

But before starting any thing we first have to see what is Cross Validation and how to use it.





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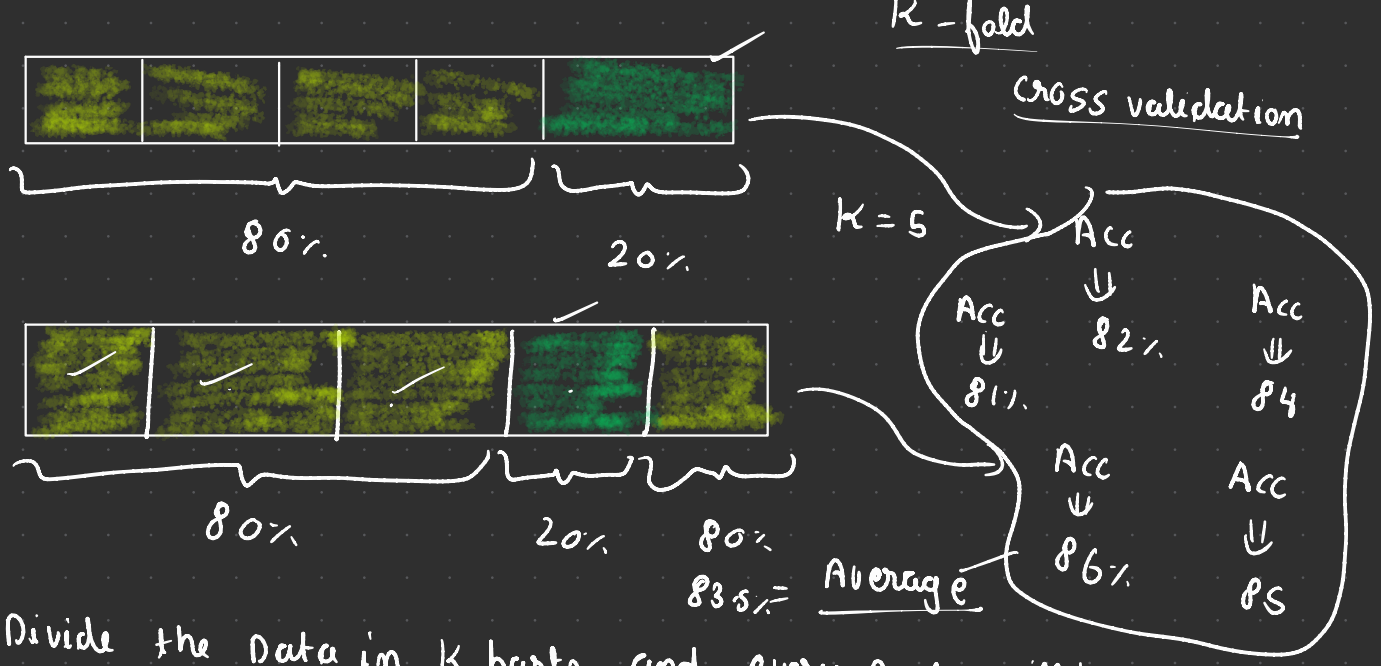
Overfit, Underfit, Poor Perform



you Divide the Data in K parts and every part will be your testing Data Remaining will be your training Data

---

hyper Parameter tuning

Ridge & Lasso Regression = α = Learning Rate

KNN ⇒ n_neighbours = [3, 5, 7, 9]

Decision Tree
- ↳ max depth
- ↳ Sample Split
- max feature

SVM
- ↳ Kernel( ),

Random forest

DBSCAN

methods
1. manual search
2. Grid search CV
3. Randomized search

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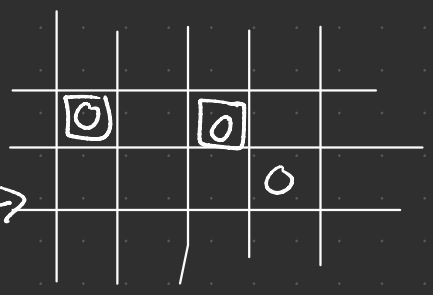
Grid Search CV

Cross validation



KNN

N = [3, 5, 7, 9, 11, 13]

weights = ["uniform", "distance"]

metric = ["manhatten", "eucledian"]

X ⟷ X

= 9 (distance, manhatten) Acc

N	weight	metric	
3	uniform	manhatten	= Acc = ?
3	distance	manhatten	= Acc = ?
3	uniform	eucledian	= Acc = ?
3	distance	eucledian	

Avg Acc ⇐ experiment ⇐

⇓

CV = 5

Random Search CV

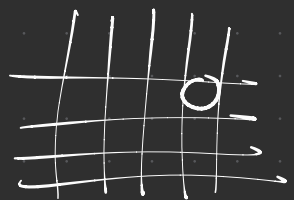
KNN

N = 6

weight = 2

metric = 2

= 6 X 2 X 2 = 24



30

XG Boost

learning Rate = 10

nEstimator = 20

maxDepth = 10

= 10 x 20 x 10

2000

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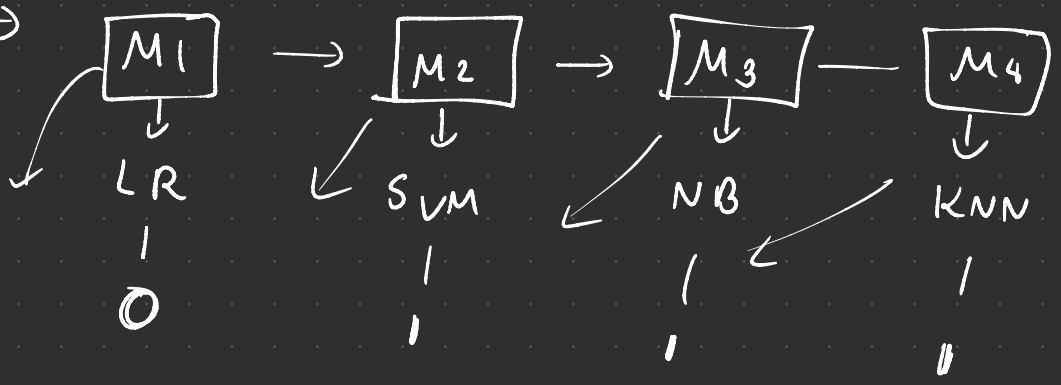
Ensemble learning

✡ we hear the crowd ✡

typing speed	Internet usage	hacker
26	12	0
58	36	1
.	.	.
,	,	,

Machine learning

Train
Test →

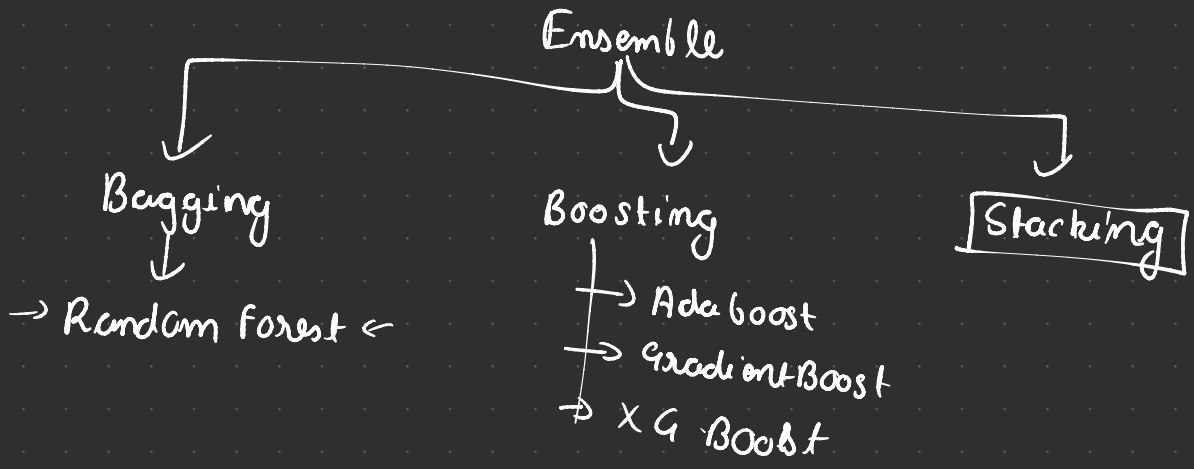


Query- TS = 56
9U = 28
hacker = ?

3y IN

Regression - mean instead of voting

Types of ensemble learning



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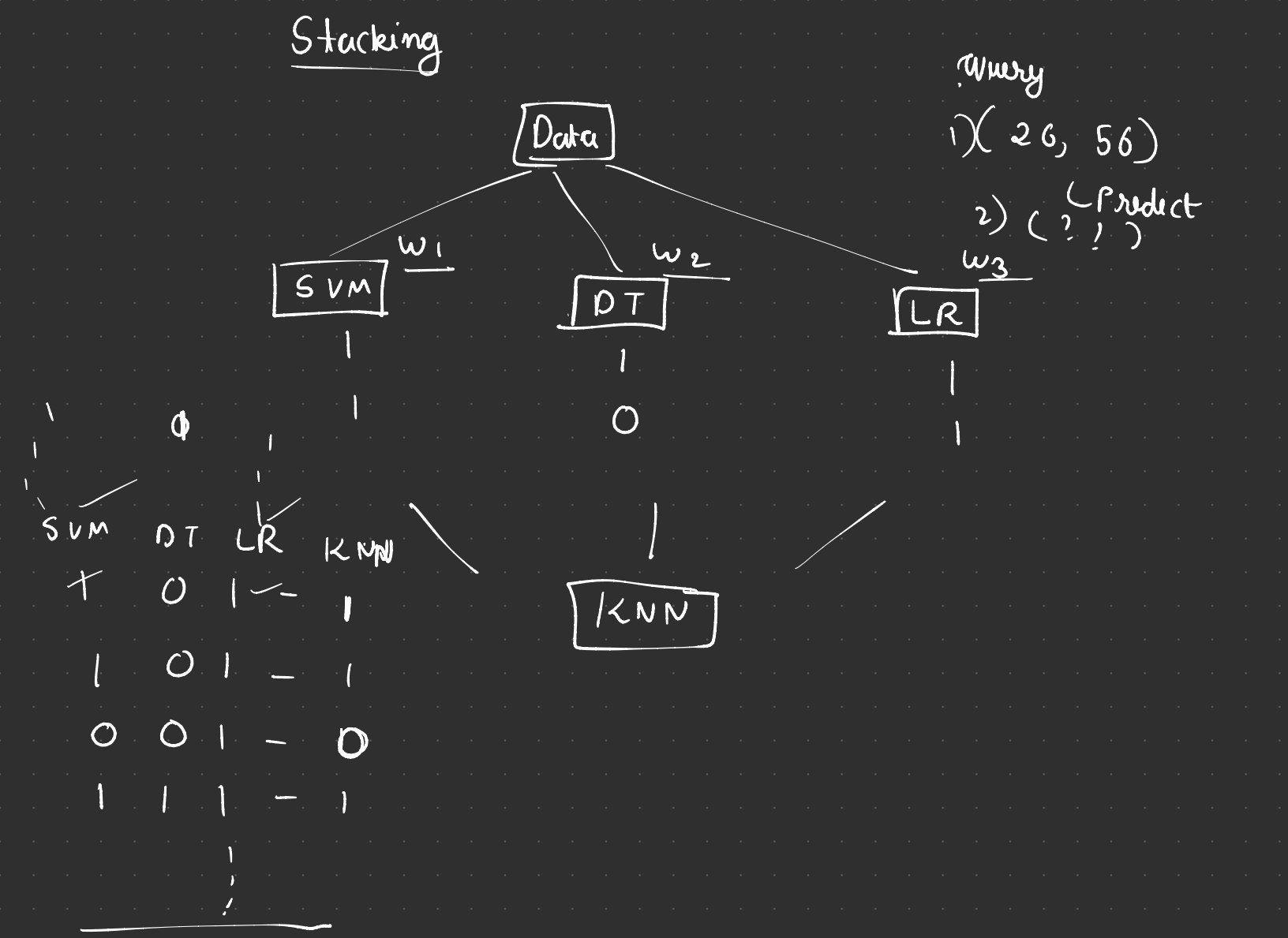
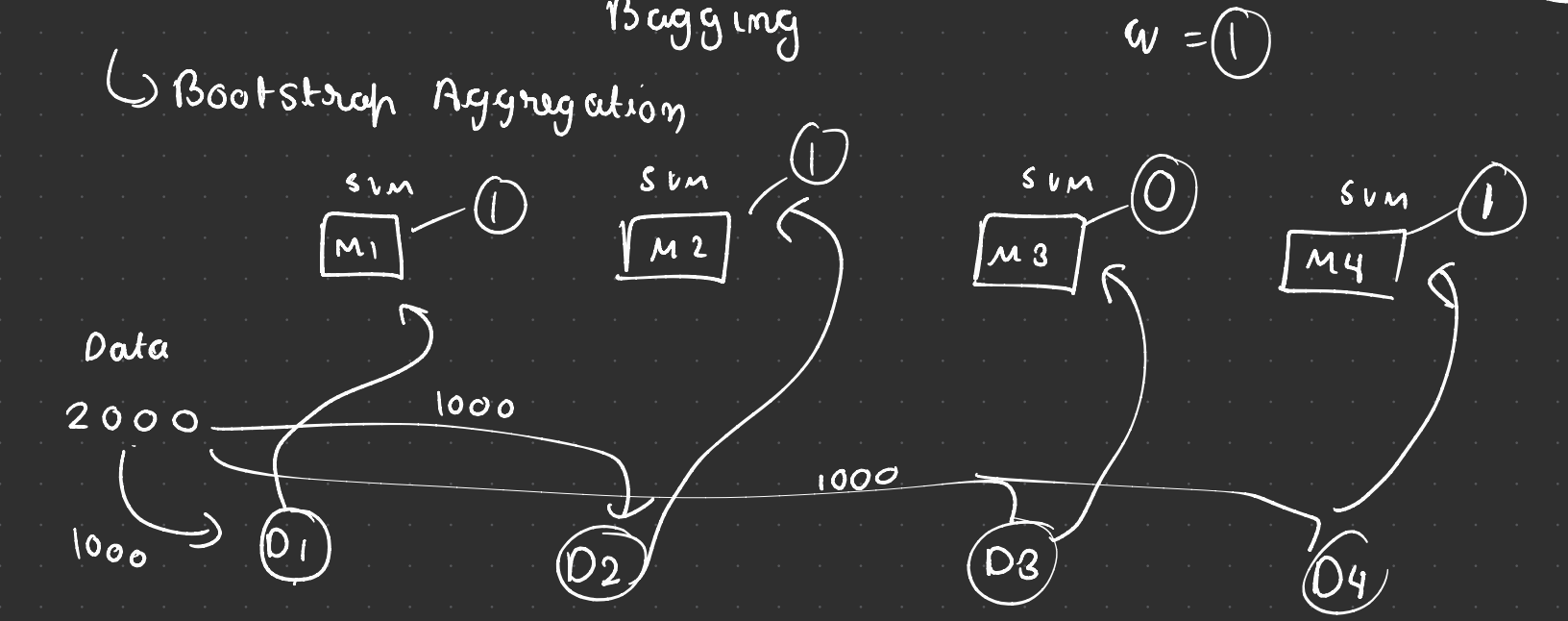
Stacking



Bagging

↳ Bootstrap Aggregation



Random forest

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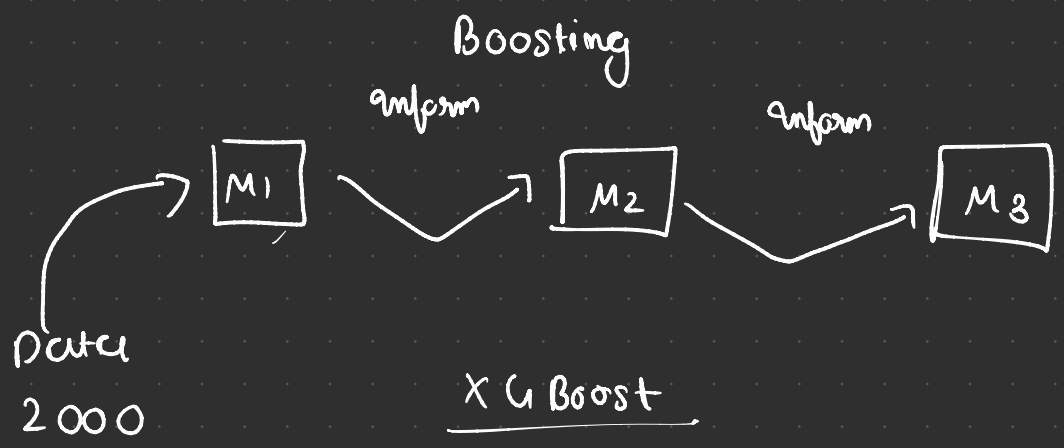
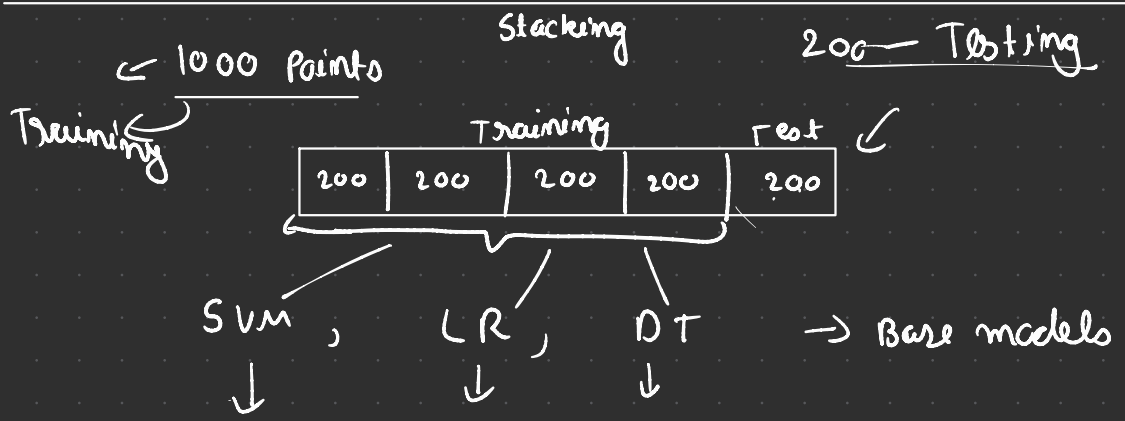


XG Boost

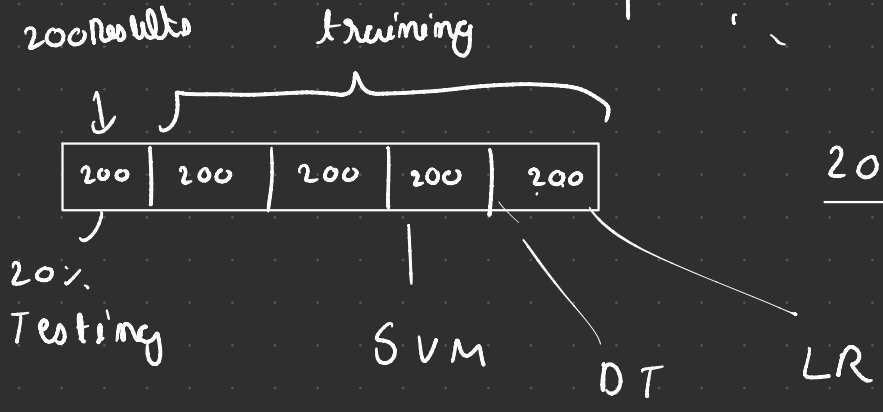


1 - 0 1 0

Meta model → KNN →

SVM_Pred,	LR_Pred,	DT_Pred	KNN
0	1	0	0
1	1	0	1
0	0	1	0
⋮	⋮		

200 Pred



Meta model - 200 + 200 = 400

↳ 1000 — Pred

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Bagging

Outlook	Temperature	Humidity	Wind	Play
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Overcast	Hot	High	Weak	Yes
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Rain	Mild	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Hot	Normal	Weak	Yes
Rain	Mild	High	Strong	No

Random forest classifier

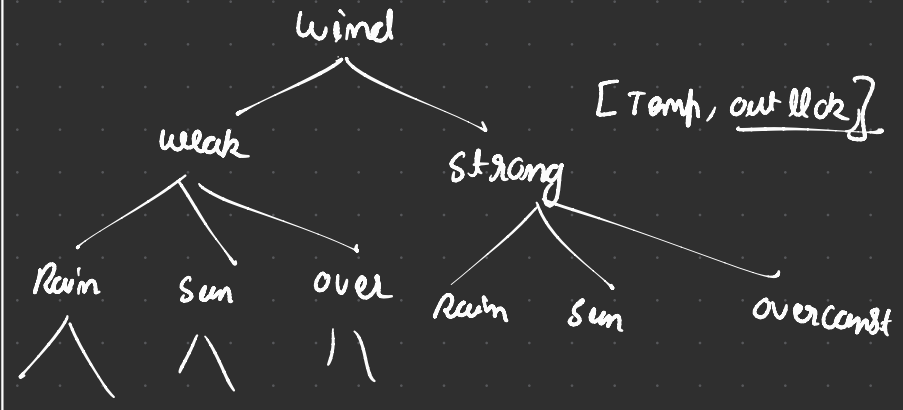
1000 DP

600 DP

100 Decision Trees

1 Tree — 600DP

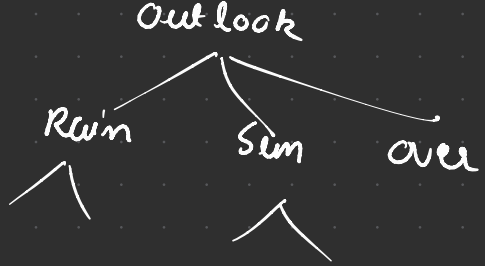
→ Root Node [Temp, wind]



[V] → Play

Tree - 2 — 600DP

[Outlook, Temp]



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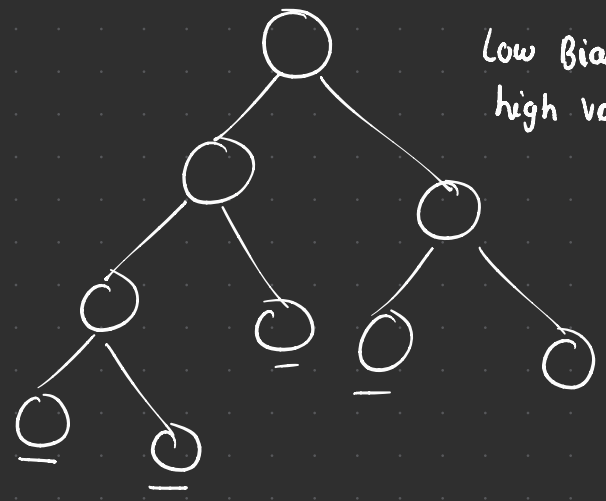
Bagging

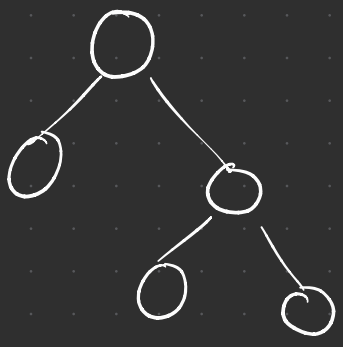


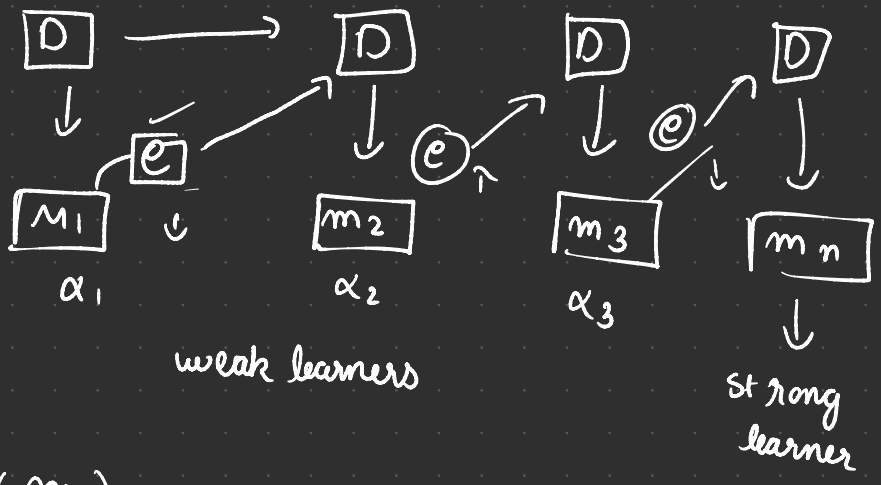
Low Bias
high variance
} → Training
Test → Low Bias
Low Variance
↓
Generalized

↳ overfitting

Boosting

Underfitting — High Bias, High variance → Generalized — Low Bias, Low variance





$$f = \alpha_1(m_1) + \alpha_2(m_2) + \alpha_3(m_3) + \alpha_n(m_n)$$

1. Ada boost
2. Gradient Boosting
3. XG Boost

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Unsupervised Learning

1)What is Unsupervised Learning?

Unsupervised Learning is a type of Machine Learning where:
The data does NOT have labels — no "right answer" is given.
The model must find patterns, structure, and relationships in the data by itself.

2) Goal of Unsupervised Learning:

Find hidden structure in the data:
Group similar data points (Clustering)
Reduce data dimensions while keeping important information (Dimensionality Reduction)
Detect outliers/anomalies
Find associations or similarities

Supervised

Data has inputs → lables
output is also available

Learn lable Prediction

Classification, Regressions

unsupervised learning

No lable → only input Data

Learn Patterns in Data

clustring, D R, Anomaly Detections

3)Common Techniques in Unsupervised Learning:

Clustering → grouping similar data points
Dimensionality Reduction → reducing number of features (PCA)
Anomaly Detection → finding outliers in data

4)Why use Unsupervised Learning?

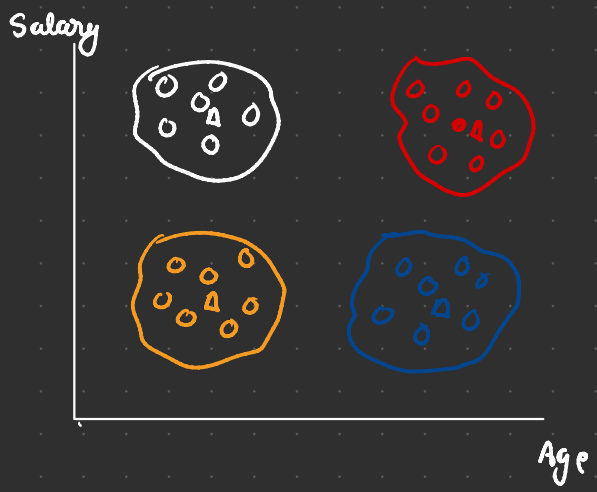
To explore and understand data structure
To discover hidden patterns
To reduce complexity of data
To find anomalies or outliers
To preprocess and improve supervised models (e.g., PCA + classifier)

5)Use Cases of Unsupervised Learning:

Customer Segmentation → group customers by behavior, spending
Market Segmentation → group markets or products
Anomaly Detection → fraud detection, network security
Recommendation Systems → find similar users/products
Image Compression → reduce image size
Medical Data → find patient groups, disease patterns
Preprocessing → use PCA to improve supervised models

Clustering

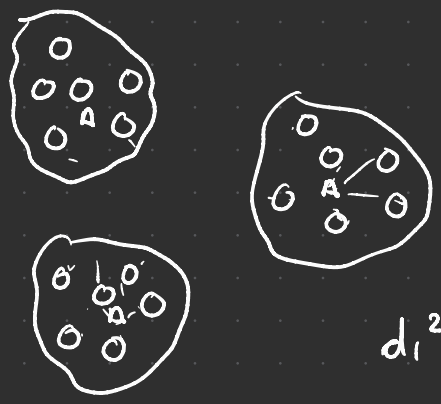
→ K means clustering

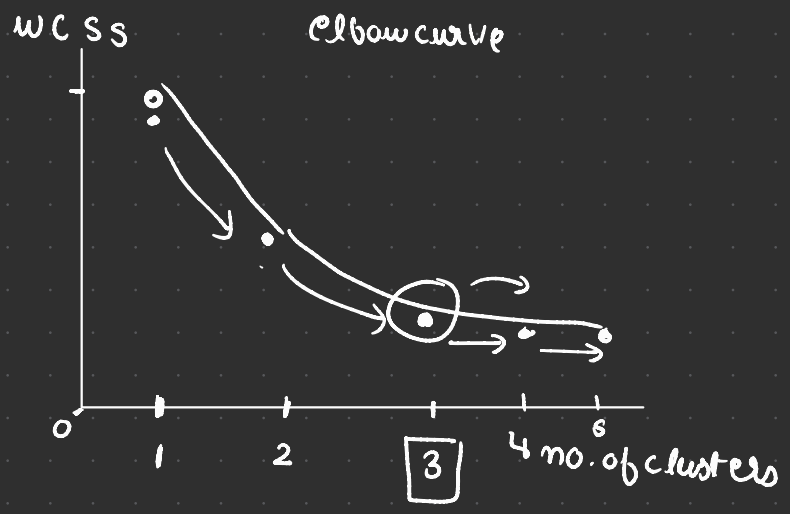


1. Decide n clusters $K = 4$
2. initialize centroids
3. Assign cluster
   ↳ Eucledian Distance
4. Reassign the center Point
5. Finish

for making the clusters and deciding K value you have to use Elbow method

wcss → within cluster sum of square distance





$$d_1^2 + d_2^2 + d_3^2 \ldots d_n^2 - \underline{wcss}$$

$$wcss_1 + wcss_2 + wcss_3 = \underline{wcss}$$

Disadvantages

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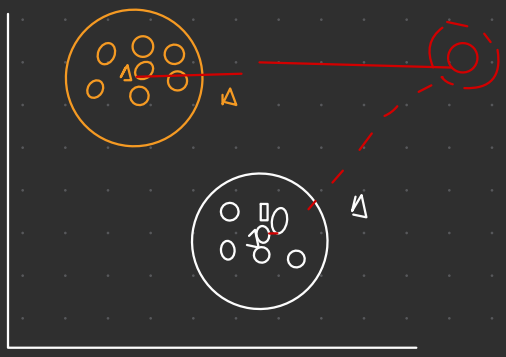
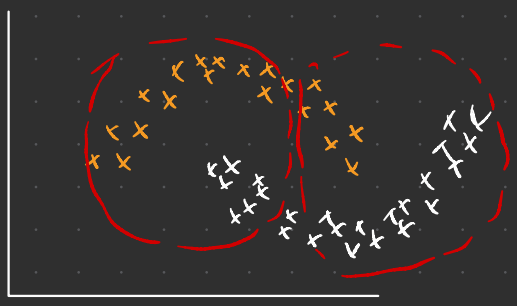
1. we have to initialize the k value
2. outliers will move the center point →
3. Non circular Data



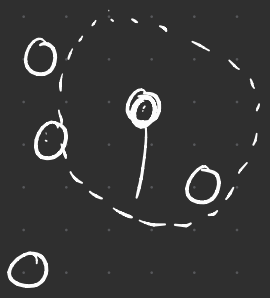


K means
↓
Centroid Based
Algo

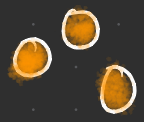
DB scan
↓
Density Based
Algo

DB SCAN

1) Non Parametric Algo
2) epsilon Distance →



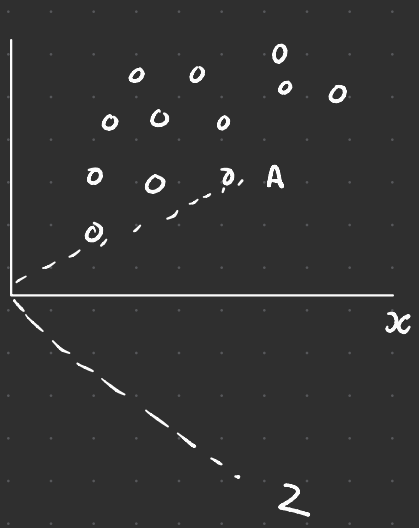
epsilon Distance = 1 unit



Dimensionality Reduction

I/P | O/P

A | Z | x | y



(x, y) 2D

(x, z, y) - 3D

100D

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Curse of Dimension

→ Very Difficult to find patterns

→ KNN, k-means, DBscan → Distance based Algo
Fail in high Dimensions

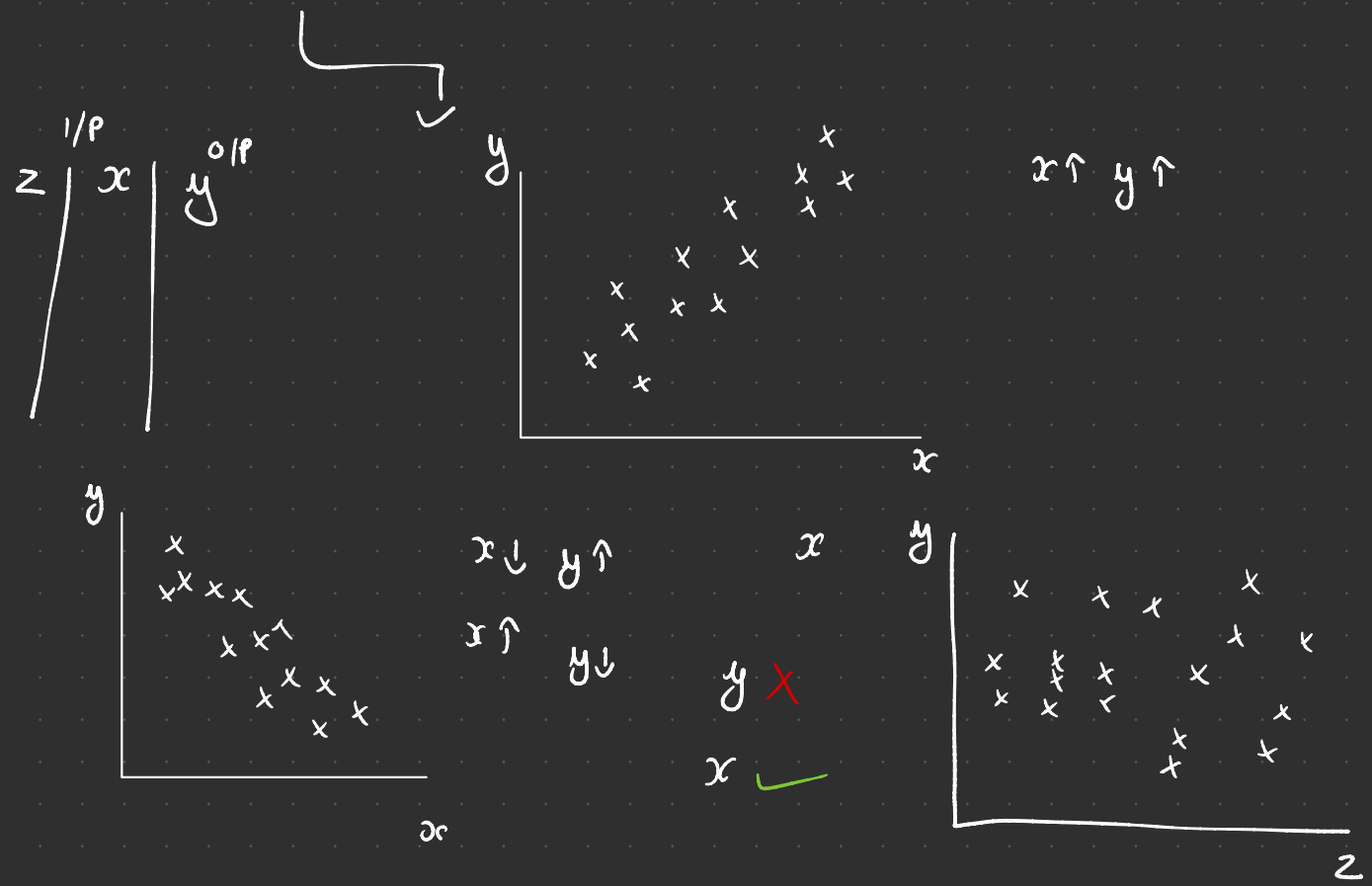
→ Storage

→ model training time

150 features ⟶ 50 features
⇓ ⇓
150D ⟶ 50D

how to achieve dimensionality Reduction

① Feature selection ② Feature extraction



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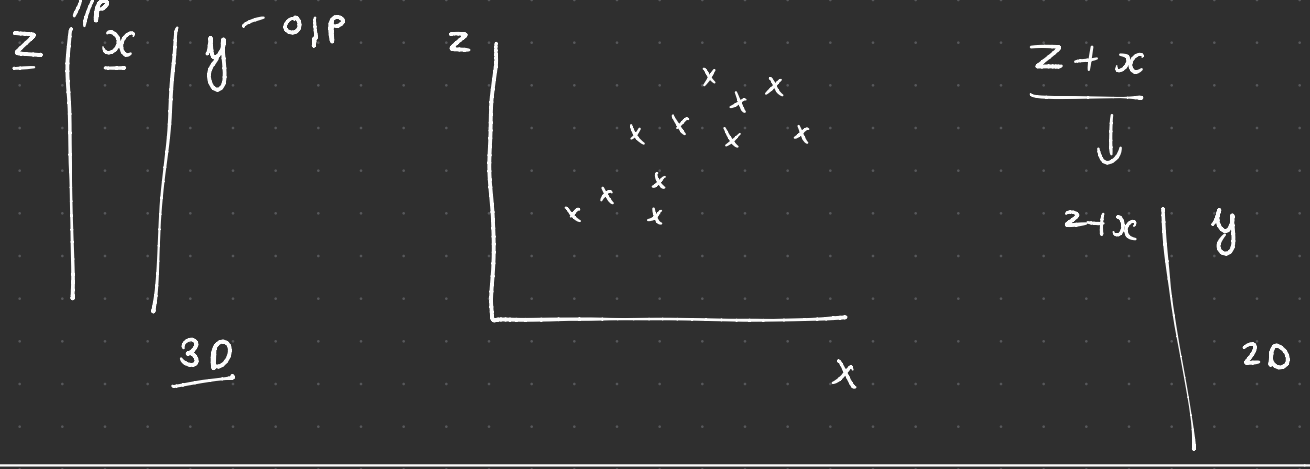
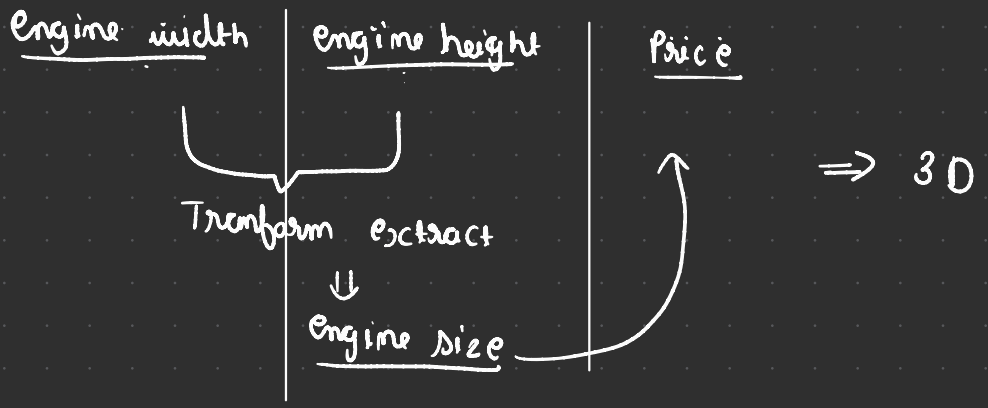
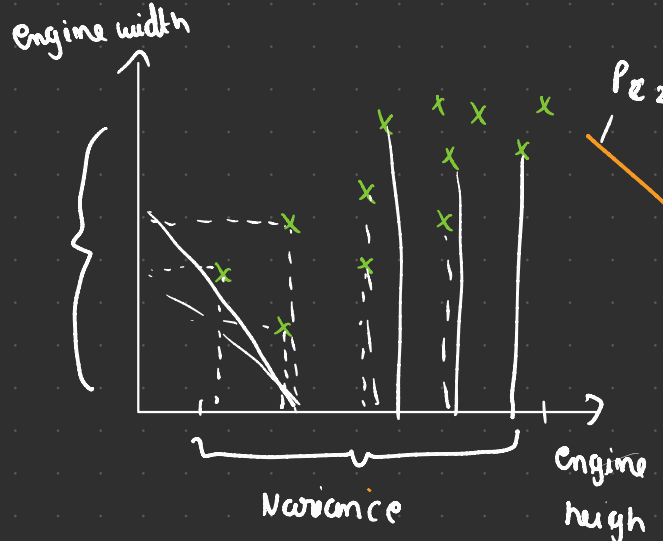
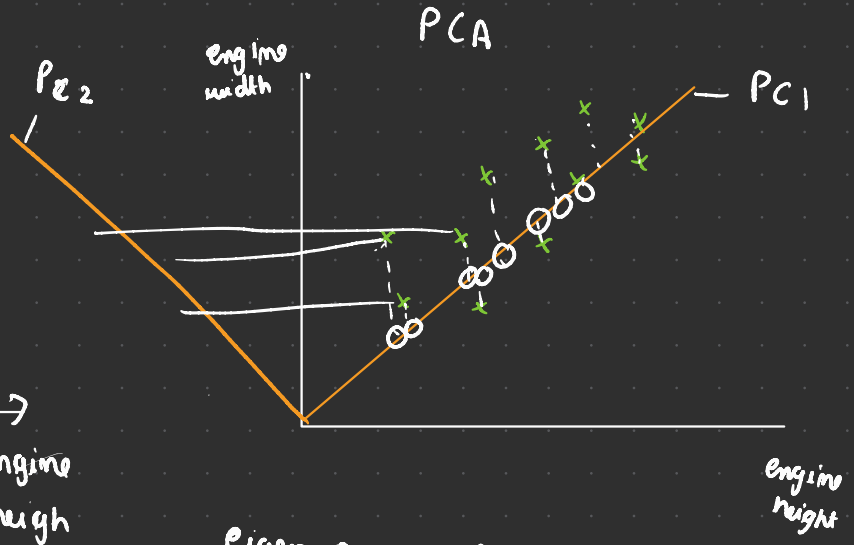




Always some amount of info. will be lost but still we can Predict the Price





Eigen Decomposition

3D — 2D

50D — 10D

✱ Yo find the PCA line where maximum variance is getting captured

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